Sequences & Series

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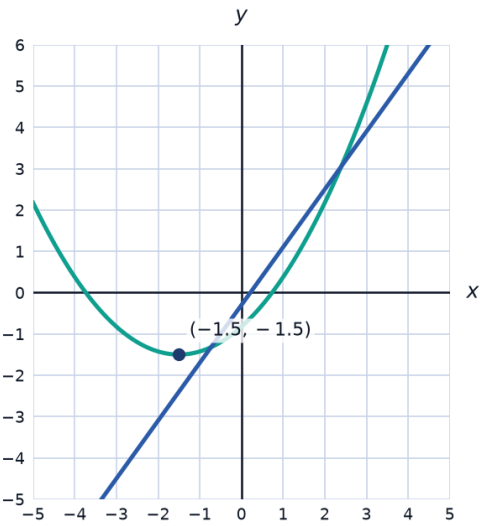
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic sequences**

For the arithmetic sequence with $a_1 = 6$ and common difference $d = 4$, find a_{12}.

2 **[Foundational]** • **Target: Arithmetic sequences**

For the arithmetic sequence with $a_1 = 20$ and $d = -3$, find a_{15}.



3 **[Foundational]** • **Target: Arithmetic sequences**

A geometric sequence has $a_1 = 3$ and each term is 2 times the previous one. Find the smallest n for which $a_n = 384$.

4 **[Foundational]** • **Target: Arithmetic series (sum formula)**

Find S_{12} for the arithmetic sequence with $a_1 = 6$, $d = 4$ (using $a_{12} = 50$ from above).

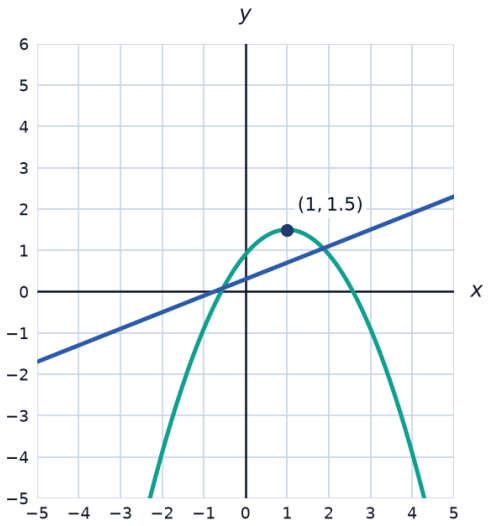
5 **[Foundational]** • **Target: Arithmetic series (sum formula)**

Find S_8 for the arithmetic sequence with $a_1 = 2$, $d = 5$.

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6 [Foundational] • Target: Geometric sequences

For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_6.



7 [Foundational] • Target: Geometric sequences

A geometric sequence begins $5, -10, 20, -40, \ldots$. Find r and a_7.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric series (finite)

Find S_6 for the geometric sequence with $a_1 = 4$, $r = 3$ (using $a_6 = 972$ from above).

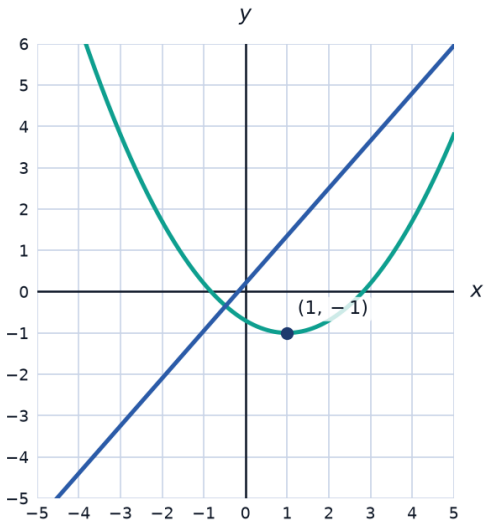
9 [Intermediate] • Target: Geometric series (finite)

Find S_5 for the geometric sequence $5, -10, 20, -40, \ldots$.

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10 [Intermediate] • Target: Infinite geometric series

Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \cdots$.



11 [Intermediate] • Target: Infinite geometric series

Explain why the infinite geometric series with $a = 5$, $r = 2$ has no finite sum.

12 [Intermediate] • Target: Arithmetic series (sum formula)

Express the repeating decimal $0.3\overline{18} = 0.3181818\ldots$ as an irreducible fraction using the infinite geometric series formula.

13 [Intermediate] • Target: Geometric sequences

The second term of a geometric sequence is 12 and the fifth term is $\frac{32}{9}$. Find the common ratio r, the first term a_1, and the sum of the infinite series S_∞.

14 [Intermediate] • Target: Geometric series (finite)

In an arithmetic sequence, the 4th term is 17 and the 10th term is 41. Determine the common difference d, the first term a_1, and the partial sum of terms from a_{11} through a_{25} inclusive.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

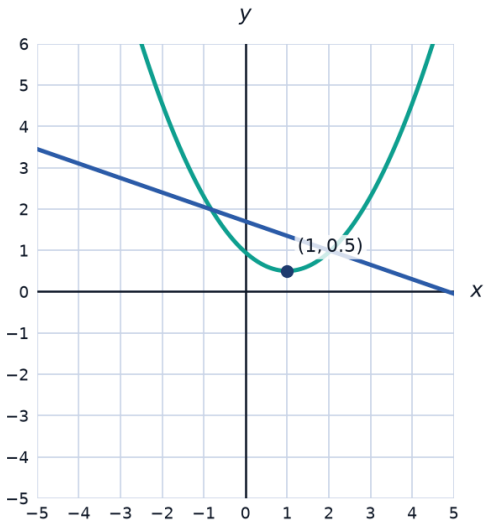
15 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 2.01 Matrices & Vectors

Evaluate the exact value of the telescoping sum: $\sum_{k=1}^{50} \left(\frac{1}{k} - \frac{1}{k+1}\right)$ and evaluate the convergent infinite series: $\sum_{n=1}^{\infty} \frac{3}{4^n}$.

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16 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 4.01 Advanced Functions & Equations

Evaluate the sum of the convergent infinite series: $24 - 12 + 6 - 3 + \cdots$



17 [Challenging] • Target: Arithmetic series (sum formula) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Evaluate the finite geometric sum: $\sum_{k=1}^{8} 3 \cdot 2^{k-1}$

18 [Challenging] • Target: Geometric sequences • Cross-Lesson: 6.01 Limits

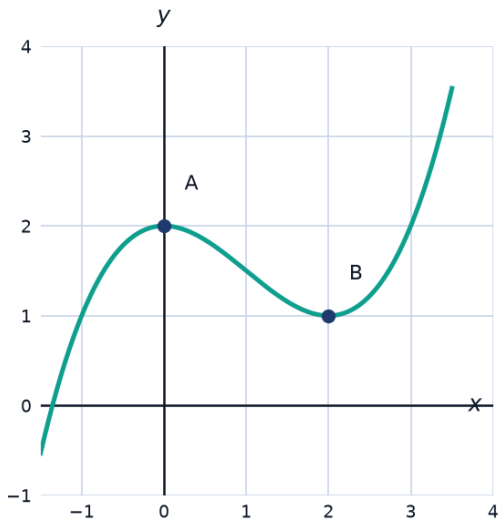
An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

19 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 1.01 Complex Numbers

An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 3$. If the sum of the first n terms is 175, find the value of n.

20 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 2.01 Matrices & Vectors

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.



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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence (a_n) and a geometric sequence (b_n) of non-zero real numbers share the same first term, so $a_1 = b_1$. The sequences also satisfy $a_5 = b_2$ and $a_6 = b_3$, with $b_2 eq b_1$. If the maximum value of the partial sum $S_N = \sum_{k=1}^{N} a_k$ of the arithmetic sequence is 153, what is the sum of the infinite geometric series $\sum_{n=1}^{\infty} b_n$?

a. 64

b. 96

c. 144

d. 192

22 [Foundational] • Target: Arithmetic series (sum formula)

An infinite geometric sequence $a_1, a_2, a_3, \ldots$ of positive real numbers has common ratio $r < 1$ and sum S. The sub-sequence of terms whose indices are of the form $3k - 1$ for positive integers k (that is, $a_2, a_5, a_8, \ldots$) has sum $\sum_{}^{\infty} a_{3k-1} = \frac{3}{13}S$. Furthermore, the infinite series formed by the squares of the terms has sum $\sum_{n=1}^{\infty} a_n^2 = \frac{9}{2}S$. What is the value of $\sum_{n=1}^{\infty} n a_n$?

a. $\frac{27}{2}$

b. $\frac{81}{4}$

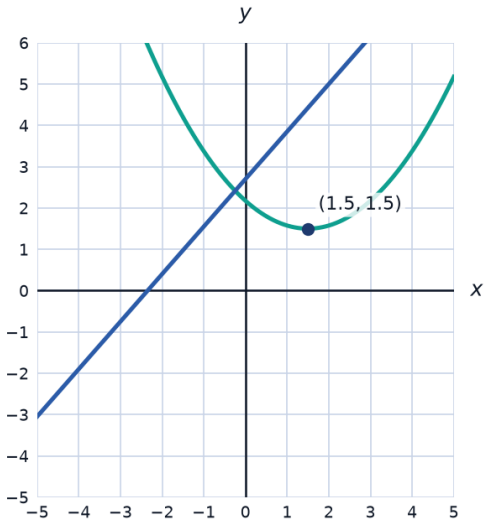
c. $\frac{9}{2}$

d. $\frac{27}{4}$

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23 [Foundational] • Target: Geometric sequences

An infinite geometric sequence of positive real numbers has first term a_1 and common ratio $r \in (0, 1)$. Let $S_1 = \sum_{n=1}^{\infty} a_n$, $S_2 = \sum_{n=1}^{\infty} a_n^2$, and $S_3 = \sum_{n=1}^{\infty} a_n^3$. If $S_1 = 4$ and $S_2 = 8$, what is the value of S_3?



a. $\frac{256}{13}$

b. $\frac{128}{13}$

c. $\frac{64}{7}$

d. 16

24 [Foundational] • Target: Geometric series (finite)

The second term of a geometric sequence of positive real numbers is k, and the fifth term is $k^2 - 20$. If the common ratio of the sequence is 2, what is the sum of the first 6 terms of the sequence?

a. 155

b. 160

c. 315

d. 630

25 [Foundational] • Target: Infinite geometric series

Does the infinite geometric series with $a_1 = 5$ and $r = -2$ converge?

a. No — $|r| \geq 1$, it diverges.

b. Yes — it converges to $\frac{a_1}{1 - r}$.

c. Yes — every geometric series converges.

d. Cannot be determined.

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26 [Foundational] • Target: Arithmetic sequences

Find the sum of the first 8 terms of the arithmetic sequence with $a_1 = -2$ and $d = -2$.

a. -72

b. -64

c. -128

d. -70

27 [Foundational] • Target: Arithmetic series (sum formula)

The first three terms of a geometric sequence of positive real numbers are $k - 1$, $k + 2$, and $3k$, in that order. What is the value of the 6th term of this sequence?

a. 24

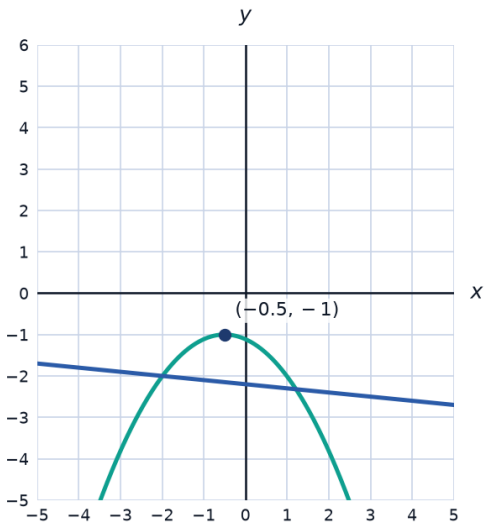
b. 48

c. 96

d. 192

28 [Foundational] • Target: Geometric sequences

In an arithmetic sequence, the 3rd term is $k + 7$, the 6th term is $3k - 1$, and the 11th term is $6k - 8$, where k is a real constant. What is the first term, a_1, of this sequence?



a. 6

b. 10

c. 19

d. 26

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Geometric series (finite)

In an arithmetic sequence, the first term is $a_1 = 4$ and the fifth term is $a_5 = 28$. What is the value of the tenth term, a_{10}?

a. 52

b. 56

c. 58

d. 64

30 [Intermediate] • Target: Infinite geometric series

The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

a. 131

b. 138

c. 140

d. 145

31 [Intermediate] • Target: Arithmetic sequences

The first four terms of an arithmetic sequence are shown below. $4, 11, 18, 25, \ldots$ What is the 31st term of the sequence?

a. 210

b. 214

c. 217

d. 221

32 [Intermediate] • Target: Arithmetic series (sum formula)

The bar chart below shows the values of the first four terms of an arithmetic sequence, $a_1, a_2, a_3,$ and a_4, where $a_1 = 6$, $a_2 = 11$, $a_3 = 16$, and $a_4 = 21$. What is the value of the 8th term, a_8, of this sequence?

a. 36

b. 41

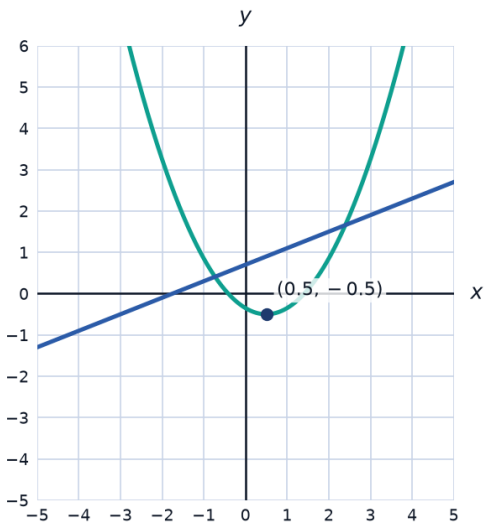
c. 46

d. 48

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33 **[Intermediate]** • **Target: Geometric sequences**

A geometric sequence has $a_1 = 3$ and common ratio $r = 3$. Find a_3.



a. 27

b. 81

c. 30

d. -27

34 **[Intermediate]** • **Target: Geometric series (finite)**

An arithmetic sequence has $a_3 = 4$ and $a_7 = -4$. Find a_1.

a. 8

b. 6

c. -8

d. 7

35 **[Intermediate]** • **Target: Infinite geometric series**

An arithmetic series has $a_1 = 5$, $d = 3$, and $S_n = 390$. Find n.

a. 15

b. 16

c. 13

d. 30

36 **[Intermediate]** • **Target: Arithmetic sequences**

Find the common difference of the arithmetic sequence $-2,\ -4,\ -6,\ -8,\ \ldots$

a. -2

b. 2

c. -1

d. -4

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37 [Intermediate] • Target: Arithmetic series (sum formula)

Find the common ratio of the geometric sequence $2,\ -6,\ 18,\ -54,\ \ldots$

a. -3

b. 3

c. -2

d. -8

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric sequences • Cross-Lesson: 6.01 Limits

Consider a geometric sequence where the first term is $a_1 = \frac{1}{2}$ and the common ratio is r. If the sum of the first three terms is $S_3 = \frac{7}{2}$, find the value of the common ratio r.

a. 2

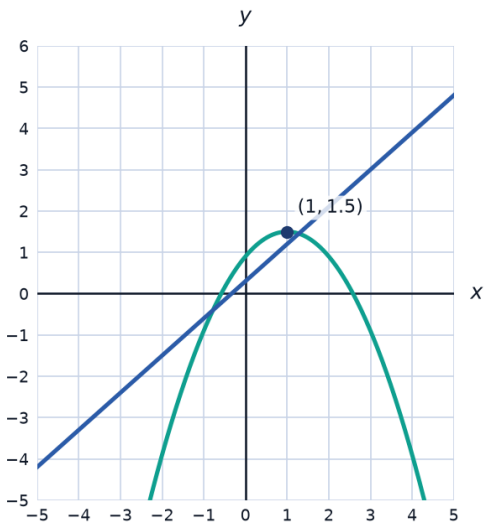
b. -2

c. 1/2

d. 3

39 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 1.01 Complex Numbers

A geometric sequence has its first term a_1 and a common ratio r. The sum of the first three terms is 26, and the sum of the first six terms is 728. If $a_1 = 2$, find the value of r.



a. 2

b. 3

c. -2

d. -3

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40 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 2.01 Matrices & Vectors

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

41 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 4.01 Advanced Functions & Equations

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. 5

c. 6

d. 7

42 [Challenging] • Target: Arithmetic series (sum formula) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

An arithmetic sequence has $a_1 = 3$ and $d = 6$. If the sum of the first n terms is 300, find n.

a. 11

b. 9

c. 5

d. 10

43 [Challenging] • Target: Geometric sequences • Cross-Lesson: 6.01 Limits

Find the sum of the first 5 terms of the arithmetic series with $a_1 = 1$ and $d = 4$.

a. 45

b. 49

c. 40

d. 50

44 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 1.01 Complex Numbers

Find the sum of the first 3 terms of the geometric series with $a_1 = -1$ and $r = 3$.

a. -13

b. -3

c. -12

d. -14

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45 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 2.01 Matrices & Vectors

If the limit $\lim_{x \to 3} \frac{x^2 + kx - 15}{x^2 - 4x + 3} = L$ exists and is a finite real number, where k and L are constants, what is the value of $k + L$?

a. 2

b. 4

c. 6

d. 8

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